

Home work 11

Task 1. For languages A, B, C, D prove the followings

- (a) $(A \cup B)C = AC \cup BC$
- (b) $A(B \cup C) = AB \cup AC$;
- (c) $A(BA)^* = (AB)^*A$;
- (d) $(A \cup B)^* = (A^*B^*)^*$;
- (e) $(A^*)^2 = A^*$;
- (f) $A^0 \cup A^1 \cup A^2 \cup A^3 = (\{e\} \cup A)^3$;
- (g) $A^*B(DA^*B \cup C)^* = (A \cup BC^*D)^*BC^*$
- (h) $(A^R)^* = (A^*)^R$;
- (i) $(A^2)^* = (A^*)^2$;
- (j) $(A \cup A^R)^* = A^* \cup (A^*)^R$;
- (k) $(A \cup B)^2 = A^2 \cup B^2$;
- (l) $(A \cap B)^2 = A^2 \cap B^2$;
- (m) $(AB)^R = A^RB^R$;
- (n) $(AB)^* = A^*B^*$;
- (o) $AA^R = A^RA$;
- (p) $(A \cap B)C = AC \cap BC$;
- (q) $A(B \cap C) = AB \cap AC$.

Task 2. Describe the languages whose regular expressions are the following

- (a) $(0 \cup 1)^*01$;
- (b) $11(0 \cup 1)^*01$;
- (c) $(0^*1^*)^*$;
- (d) $(01^*)^*$;

- (e) $(0 + 10)^*(\varepsilon + 1)$;
- (f) $(0 + 10 + 110)^*(\varepsilon + 1 + 11)$;
- (g) $0(0 + 1)^*1 + 1(0 + 1)^*0$;
- (h) $1^*(01^*01^*)^*$;

Task 3. Write regular expression for the following languages:

- (a) $L = \{x \in \{0, 1\}^* : x \text{ is started with } 000\}$;
- (b) $L = \{x \in \{0, 1\}^* : x \text{ is ended with } 111\}$;
- (c) $L = \{x \in \{0, 1\}^* : x \text{ contains the substring } 011\}$;
- (d) $L = \{x \in \{0, 1\}^* : x \text{ is started with } 000 \text{ or ended with } 111\}$;
- (e) $L = \{x \in \{0, 1\}^* : x \text{ is started with } 000 \text{ and ended with } 111\}$;
- (f) $L = \{x \in \{0, 1\}^* : x \text{ is started with } 000, \text{ ended with } 111 \text{ and contains the substring } 011\}$;
- (g) $L = \{x \in \{0, 1\}^* : \text{the first and last symbols of } x \text{ are same}\}$;
- (h) $L = \{x \in \{0, 1\}^* : \text{number of symbol } 0 \text{ in } x \text{ is even}\}$;
- (i) $L = \{x \in \{0, 1\}^* : x \text{ does not contain substring } 010\}$;
- (j) $L = \{x \in \{0, 1\}^* : x \text{ is started with } 000 \text{ and length of } x \text{ is even}\}$;

Task 3. Find the first 3 shortest strings in the following regular expressions:

- (a) $10 + (0 + 11)0^*1$;
- (b) $(00 + 11 + (01 + 10)(00 + 11)^*(01 + 10))^*$;
- (c) $((00 + 11)^* + (001 + 110)^*)^*$;
- (d) $(a + bc^*d)^*bc^*$;
- (e) $(ab + bc^*d)ab^*(c + dc)$.

Task 4. Find the graph representations for the following regular expressions

- (a) $01 + (0 + 11)0^*1$;
- (b) $(00 + 10)(101)^* + 01$;
- (c) $(00 + 10)^*101(0 + 1)^*$;

(d) $(a + bc^*d)^*bc^*$;

(e) $(ab + bc^*d)ab^*(c + dc)$.

Task 5. Write regular expressions whose graph representations are the following:

